

CURRENT FINAL TERM BT503

EFFORT BY: WARRIORS + ASHAM

1. Consequences of OECD 2001 about global environment 3

The OECD (2001) concluded that the industrial use of biotechnology commonly leads to increasingly environmentally harmonious processes and additionally results in lowered operating and/or capital costs.

2. What are qualities of pollutant that are intended target in bio processing? 5

The intended target of the bio-processing must generally be both susceptible and available to biological attack, in aqueous solution, or at least in contact with water, and within a low to medium toxicity range. For land based applications, the soil types best suited to biotechnological interventions are sands and gravels. Moreover, nutrient availability, oxygenation and the presence of other contaminants can all play a role too.

3. Extremophile 6

In general the use of biotechnology for environmental management relies on mesophilic microorganisms. Some microbial species tolerate extreme conditions, like high salinity, pressures and temperatures, useful for biotech applications that tolerance to these conditions

The Archaea

They are extreme thermophiles, extreme halophiles, and methanogens, species that tolerate high levels of ionizing radiation, pH, or high pressure. Extremophiles provide way of developing alternative routes to many conventional chemicals or material

Much of the interest centers on the extremophile enzymes, the so-called extremozymes. The promise of extremozymes lies in their ability to remain functional when other enzymes cannot.

4. Thermopolis found 7

A thermophile is an organism, type of extremophile that thrives at temperatures above 45°C. Many thermophiles are archaea, though they can be bacteria. Hot-springs, deep sea-vents, geothermal fluids, etc.

5. Acidophile 8

Acidophiles thrive in the conditions of low pH, typically below. Protective molecules present in their cell walls, membranes or outer cell coatings to exclude acids

6. Endocrine disrupter 10

There are chemicals which resist degradation. Partly due to lack of organisms with degradative abilities, or worse due to microbial activities that change them to more toxic form

7. Difference between transductions, transformation, plasmid and conjugation. 12

Transformation: In transformation, a bacterium takes piece of DNA floating in its environment.

Transduction: In transduction, DNA is accidentally moved from one bacterium to another by a virus.

Conjugation: In conjugation, DNA is transferred between bacteria through a tube between cells.

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Plasmid: A plasmid is a small, extra chromosomal DNA molecule within a cell that is physically separated from chromosomal DNA and can replicate independently.

8. Catabolic expansion? 14

Catabolic expansion: Improving microbe's ability to degrade a contaminant by culturing the bacteria in growth medium in which the contaminant is supplied as an essential part of the nutrition. Only bacteria which have undergone a mutation enabling them to utilize this food source will be able to survive.

9. Fundamental requirements of all cloning techniques 16

- ♥ The enzymes, solutions and equipment necessary to perform the procedures
- ♥ The desired piece of DNA to be transferred
- ♥ a cloning vector
- ♥ The recipient cell.

10.cDNA libraries 18

cDNA libraries In eukaryotes, the first product of transcription from DNA is not messenger RNA but heterogeneous nuclear RNA (hnRNA). This is mRNA prior to the removal of all the non-coding sections or introns which are discarded during the processing to produce the mature mRNA. cDNA is DNA which has been artificially made using the mature mRNA as a template

11.Blotting northern, Western 21

Northern blotting detects a specific RNA sequence from an RNA sample.

Southern blotting detects a specific DNA sequence from a DNA sample.

Western blotting detects a specific protein from a protein sample by using Agarose /formaldehyde gel

12.Transgenic plant 24

“A transgenic plant is a modified organism where genes are transferred from one organism to another through genetic engineering techniques”.

13.Ways to improve crop quality 24

Crop improvement refers to the ways of developing and breeding of crop varieties which are better than the pre-existing cultivars in a number of characters like taste, height, colour, etc. for a better yield. Many methods used to improve crop performance. Others involve improvements in farm practices, irrigation, and drainage, and herbicide, pesticide and fertiliser use.

14.T DNA 26

The transfer DNA is the transferred DNA of the tumor-inducing (Ti) plasmid of bacteria such as *Agrobacterium tumefaciens* and *Agrobacterium rhizogenes*. The T-DNA is transferred from bacterium to host plant's nuclear DNA genome.

15.How plants protect themselves from TMV 28

Introduction of the genes expressing antibodies to the coat protein of Tobacco Mosaic Virus (TMV) by *A. tumefaciens*. Expression of these in the plant leads to complete immunity against TMV.

16.Quorum sensing 29

Bacteria communicate with each other by way of small diffusible molecules such as the Nacylhomoserine lactones (AHLs)

17.Natural processes to remove the pathogens organism. 29

Natural: Maturation Ponds, Land treatment

Maturation ponds: penetration of solar UV radiation and unfavorable environmental conditions causes a high mortality of the pathogens.

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Land treatment: The unfavorable environmental conditions in the soil favor the mortality of the pathogens. Chemical products are not needed. Requires large areas

18.What are cynophyta? 34

Cynophyta, a type of phytoplankton, is one particularly powerful bioindicator which is known to indicate rapid eutrophication of water bodies such as reservoirs, lakes, etc. via the creation of bloom formations.

19.Bio indicators 33

Bioindicators are living organisms such as plants, planktons, animals, and microbes, which are utilized to screen the health of the natural ecosystem in the environment.

20.How plants are modified as bioindicator 35

The presence or absence of some specific plants or other vegetation provides ample information about environmental health. e.g., Lichens generally found on the trunks of trees and rocks, react to ecological changes in forests, including changes in the structure of the forest, air quality, and climate

21.Phytoplankton 41

Phytoplanktons, also known as microalgae, are similar to terrestrial plants in that they contain chlorophyll and require daylight to live and develop. Most are light and swim in the upper portion of the sea, where light infiltrates the water

22.Zoo planktons 43

The word "zooplankton" is derived from the Greek zoon, meaning "animal", and planktos, meaning "wanderer" or "drifter".

Zooplanktons are microscopic animals living near to the surface of the water body. They are poor swimmers, instead relying on tides and currents as a transport mechanism. They feed upon phytoplankton, bacterioplanktons, or detritus (i.e. marine snow). Zooplanktons constitute a vital food source for fish. They also play an important role as Bioindicators and help to evaluate the level of water pollution.

23.Environment pollution? 46

Any change in our environment that has harmful effects on plants, animals and human beings is called environmental pollution.

24.Role of filter medium in boiler also name the compound of the filter medium. 60

The first methods to be developed that consists of a relatively large vessel or container, typically made of cast concrete, metal or durable plastic, which holds a filter medium of organic material such as peat, heather, bark chips and the like. The gas to be treated is forced, or drawn, through the filter.

The medium offers good water holding capacity and soluble chemicals within the waste gas dissolve into the film of moisture around the matrix. Bacteria, and other micro-organisms present, degrade components of the resultant solution, thereby bringing about the desired effect. The medium itself provides physical support for microbial growth, with a large surface area to volume ratio, high in internal void spaces and rich in nutrients to stimulate and sustain bacterial activity. Bio filters need to be watered sufficiently to maintain optimum internal conditions, but waterlogging is to be avoided as this leads to compaction, and hence, reduced efficiency. Properly maintained, bio filters can reduce odor release by 95% or more.

25.Biofilters and its phases 60

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Biofiltration is a pollution control technique using a bioreactor containing living material to capture and biologically degrade pollutants.

This bio-filter is a three-phase reactor composed of:

- ♥ Solid phase: consists of a support medium and biofilms
- ♥ Liquid phase: consists of the liquid in permanent flow through the porous medium
- ♥ Gas phase: formed by artificial aeration and by the gaseous by-products of the biological activity.

26.Five advantages of bio filter 60

- ♥ Low investment
- ♥ Simple construction
- ♥ Effective removal of biologically degradable components
- ♥ Low pressure drop
- ♥ Little waste water
- ♥ Little waste material

27.Bio scrubber 62

The bio scrubber is not itself truly a biological treatment system, but rather a highly efficient method of removing odour components by dissolving them. Most appropriate for hydrophilic compounds like acetone or methanol.

28.Advantages of trickling filter over biofilter 62

The efficiency of the bio trickling filter is broadly similar to the bio filter it can deal with higher concentrations of contaminant and has smaller foot-print than a biofilter of the same throughput capacity. Although trickling filters are more easily operated and consume less energy than activated sludge processes, they have lower removal efficiency for solids and organic matter, they are more sensitive to low air temperatures

29.Difference between adsorption and incineration 64

Absorption is a chemical or physical phenomenon in which the molecules, atoms and ions of the substance getting absorbed enters into material in which it is taken up.

Incineration refers to the destruction of material such as waste. Incineration is the high-temperature burning of a waste

30.Biomimetic 78

“Biomimetics is an interdisciplinary field in which principles from engineering, chemistry and biology is applied to the synthesis of materials, synthetic systems or machines that have functions that mimic biological processes.”

31.Waste water treatment 81

Wastewater treatment is the process of converting wastewater water that is no longer needed or is no longer suitable for use into bilge water that can be discharged back into the environment. It's formed by a number of activities including bathing, washing, using the toilet, and rainwater runoff.

32.Maturation ponds 84

The main objective is the removal of pathogenic organisms. Adverse environmental conditions, such as UV radiation, high pH, high DO, lower temperature, lack of nutrients, etc. Maturation ponds are a post-treatment stage for BOD-removal processes.

33.Advantages of conventional biological sludge 111

Advantages of Conventional Dispersed Plug Flow Process

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- ♥ Allows smaller volume than CSTR
- ♥ Flexible operation, zone aeration, step feeding options, accommodates anoxic and aerobic processes with single biomass for biological nutrient removal
- ♥ Less aeration than CSTR
- ♥ Can be used for sludge thickening and dewatering
- ♥ Low land requirements
- ♥ Ease of installation

34. Phases of aerated biofilters 118

Solid phase: consists of a support medium and biofilms

Liquid phase: consists of the liquid in permanent flow through the porous medium

Gas phase: formed by artificial aeration and by the gaseous by-products of the biological activity.

35. Biodisc 119

Bio-disc technology is an extensive wastewater treatment process involving aerobic biological digestion with fixed biomass.

36. Main purpose of bio disc 119

- ♥ to serve as the surface for microbial film growth
- ♥ to promote the contact between the microbial film and the sewage
- ♥ To maintain the biomass that detached from the discs in suspension in the liquid
- ♥ To promote the aeration of the sewage that is agree to the disc and the sewage immersed in the liquid.

37. Stages of Sludge waste management 128

The main stages in sludge management:

- ♥ Thickening: removal of water (volume reduction)
- ♥ Stabilisation: removal of organic matter (mass reduction)
- ♥ Conditioning: preparation for dewatering
- ♥ Dewatering: removal of water (volume reduction)
- ♥ Disinfection: removal of pathogenic organisms
- ♥ Final disposal: final destination

38. Difference between physical and mechanical sludge dewatering process 128

Sludge is separated in a solid and a liquid part with the help of a physical process such as filtration, squeezing, centrifugal separation or compaction.

Mechanical dewatering is mostly applied for the treatment of residual sludge in large scale centralized wastewater treatment plant

39. Sludge dewatering 130

Sludge dewatering is the practice of minimizing waste by volume to prepare for its effective disposal.

40. dewatering main objective 130

The main objectives:

- ♥ Reduction of the transportation costs
- ♥ Improvement of the handling conditions
- ♥ Increase of the calorific value of the sludge (for incineration)
- ♥ Reduction in the volume for disposal

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- ♥ Reduction in the production of leachate when the sludge is disposed of in landfills

41.natural and mechanical dewatering difference 130

Sludge dewatering can be done by natural or mechanised processes. Natural processes use evaporation and percolation as the main water removal mechanisms, thus requiring more time for dewatering. Although simpler and cheaper to operate, they need larger areas and volumes for installation. In contrast, the mechanized processes are based on mechanisms such as filtration, compaction, or centrifugation to accelerate dewatering, resulting in compact and sophisticated units, from an operational and maintenance point of view.

42.Mention the most common metallic coagulants for sludge dewatering 130

Aluminium sulphate • ferric chloride • ferrous sulphate • ferric sulphate • quicklime/ hydrated lime

43.Advantage and disadvantage of incineration 135

Advantages: • Drastic volume reduction • Sterilization

Disadvantages: • High costs • Ash disposal • Atmospheric pollution

44.Disadvantage of in situ and ex situ? 138

Ex situ Disadvantages: * Additional transport costs, * increased likelihood of spillage

In situ disadvantages: Stringent requirement for thorough site investigation and survey. Since reaction conditions are not readily controlled, the supposed process 'optimisation' may, in practice, be less than optimum

45.Characteristics of intensive technology 139

Intensive technologies can be characterized as fast acting, high intervention strategies, with a heavy demand for resources and high initiation, running and support costs. Their key factors are a fast response and low treatment time, which makes them excellent for heavy contamination conditions. E.g., Soil washing and thermal treatments

46.Bioremediation and its types 141

Bioremediation is the use of microbes to clean up contaminated soil and groundwater. Microbes are very small organisms, such as bacteria, that live naturally in the environment.

Three major routes of Bioremediation:

- ♥ Mineralization
- ♥ Co-metabolism
- ♥ Immobilization.

47.Factors effect bioremediation 142

Environmental Factors:

Temperature: 0 and 50 °C. Most efficient: 20–30 °C

PH: pH range of 5.0-9.0, however, 6.0-8.0 being most efficient.

Soil type: sands and gravels are the most suitable soil types for bioremediation

Other factors:

Nutrient availability, oxygenation, presence of other inhibitory contaminants, the groundwater runs off, contaminants that are present, concentration of contaminants and place where they are biodegradable, etc.

48.Bio augmentation and bio enhancement 143

Bioenhancement concentrates solely on the existing micro-fauna, stimulating their activity by the manipulation of local environmental conditions.

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Bioaugmentation requires the deliberate introduction of selected microbes to bring about the required clean-up.

49. Phyto-remediation 150

“Direct in situ use of living green plants for the treatment of contaminated soil, sludges or ground water that remove degradation or containment of the pollutants present”

50. Photovoltlization 153

It involves the uptake of the contaminants by plants and their release into the atmosphere, typically in a modified form. Relies on the transpiration pull of fast-growing trees, which accelerates the uptake of the pollutants in ground water solution, which are then transformed and released through the leaves

E.g., genetically modified variety of the Yellow Poplar, *Liriodendron tulipifera* Engineered by the introduction of mercuric reductase gene (mer A). This confers the ability to tolerate higher mercury concentrations, and removal of the pollutant from the soil and volatilize it. In a number of studies, poplars have been shown to be able to volatilize around 90% of the TCE (Trichloroethylene) they take up.

51. Phytodegradation and its forms 153

Phytodegradation involves the degradation of organic contaminants directly, through the release of enzymes from roots, or through metabolic activities within plant tissues.

Forms are:

- ♥ Internally, having first been taken up by the plants,
- ♥ Externally, using enzymes secreted by them

52. Enlist sources of solid waste 162

- ♥ Organic waste: Waste from preparation of food, market places, etc.
- ♥ Combustibles: Paper, wood, dried leaves, packaging, etc. (high organic and low moisture content)
- ♥ Different categories of solid waste include:
 - ♥ Non-combustibles: Metal, tin cans, bottles, stones, etc.
 - ♥ Ashes/dust: Residue from fires used for cooking.
 - ♥ Bulky waste: Tree branches, tyres, etc.
- ♥ Different categories of solid waste include:
 - ♥ Dead animals: Carcasses of domestic animals and livestock.
 - ♥ Hazardous waste: Oil, battery acid, medical waste.
 - ♥ Construction waste: Roofing, rubble, broken concrete, etc.

53. Fermentation produce solution of ethanol in water 176

We can be further treated to produce fuel-grade ethanol by subsequent simple distillation, to 95% ethanol, or to the anhydrous form by azeotropic co-distillation using a solvent.

54. Advantages of biodiesel 179

Biodiesel is derived mostly from vegetable oils

Transesterification

It can be used directly, in unmodified engines, with the additional bonus that it can perform as a single, pure fuel or as part of a mix with its traditional counterpart. Biodiesel exhaust is generally less harmful to both human health and the planet.

55. Advantages of biofuels 180

Enormous energy demand particularly in industrialized countries

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The production of biofuels has accordingly burgeoned in recent years. They are driven by the twin pressures of environmental and economic necessity. The rush towards biofuels could have larger social and environmental consequences. UN reported growing evidence that food prices were being driven up in poorer nations of the world as oil rich crops were grown for fuel.

56.Bio pesticides? 183

Biopesticides are certain types of pesticides derived from such natural materials as animals, plants, bacteria, and certain minerals. For example, canola oil and baking soda

57.Association of fungi with plants 186

Mycorrhizae: Associations of fungi with roots of vascular plants quite common and in some cases be very beneficial to the plant. They may be external, ectomycorrhizal, or internal, endomycorrhizal.

58.Minimum substrate concentration 200

Biodegradation of synthetic organic compounds is strongly affected by S_{min} , The lower threshold concentration below which reaction speed is insufficient to supply the organism with sufficient energy for net growth. Many hazardous organic compounds have drinking-water standards that are very low.

Such concentrations are frequently less than S_{min} While a given compound may be easily biodegradable at higher concentrations.

59.Main chemicals characteristics of chlorinated pesticides that lead to problems 204

Two characteristics of chlorinated pesticides that led to great harm are their resistance to biodegradation and their hydrophobicity. Chlorinated pesticides are taken up by plankton and concentrated.

60.Three Sources of dioxins 220

Dioxins are mainly byproducts of industrial practices. They are produced through a variety of incineration processes, including improper municipal waste incineration and burning of trash, and can be released into the air during natural processes, such as forest fires and volcanoes.

61.dioxin effects on human 220

Human responses include chloracne, weight loss, insomnia, liver dysfunction, spontaneous abortion, and possibly cancer.

62.Biosensor 228

A biosensor is an analytical device, used for the detection of a chemical substance that combines a biological component with a physicochemical detector

63.With the help of named example discuss how whole organism can be used to measure potential biological toxicity of water and soil 230

Much effort use to develop and use different bio assays and biosensors for toxicity evaluation of water samples. Whole organisms to measure the potential biological toxicity of a water or soil sample. Microtox® or ToxAlert® toxicity assays use the luminous *Vibrio fischeri* bacteria to detect the toxicity of substances. For quick ecotoxicity investigation, use Cellsense®, an amperometric sensor that contains *Escherichia coli* cells.

64.What are pcbs? which biosensors detect them 234

Polychlorinated biphenyls (PCBs) are ubiquitous environmental pollutants. They are widely used as industrial chemicals, particularly as dielectric fluids in electrical transformers and capacitors. Though the production of PCBs has been banned, these are still present in the environment. The high toxicity represents a risk for public health.

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Different biosensor determines PCBs in the environment. DNA biosensor with chronopotentiometric detection, various immunosensors with fluorescence, SPR, and electrochemical detection principles use for this determination.

Warriors